HARBORTECH SOLUTIONS

Cloud Operations Support Practice

IT 4560 · Week 06 Lab Submission

Do We Need a Server? Compute Model Recommendation

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Tuesday, October 13, 2026

SECTION 2 | EVIDENCE REVIEW AND ANALYSIS

Response 1 | Daily report: compute model and workload evidence

A daily report is a good fit for AWS Lambda because it runs briefly, only once each morning, and does not need to store local state between runs. This avoids paying to keep an EC2 instance idle.

Response 2 | API Gateway and Lambda: request flow and responsibilities

API Gateway receives API requests and routes them to Lambda, which runs the application code. HarborTech still manages the code, error handling, and integrations with other services.

Response 3 | Continuously running workload: VM, container, or Lambda

Lambda is less suitable because the application needs a custom OS package and host-level troubleshooting access. A VM may be better for full OS control, while containers may work if the package is supported. Check OS requirements and access needs first.

SECTION 3 | COMPUTE MODEL RECOMMENDATION

HarborTech internal recommendation | TKT-2026-0006

The current EC2 instance runs 24/7 but has low utilization, averaging only 2% CPU. The reporting task runs once each morning for about 4.5 minutes, while the API receives intermittent requests throughout the day. I recommend moving both workloads to serverless services because they only need to run when triggered.

For reporting, EventBridge Scheduler would trigger AWS Lambda each morning. For the API, API Gateway would handle incoming requests and route them to Lambda. AWS would manage the underlying servers, operating system, scaling, and infrastructure maintenance. HarborTech would still manage the application code, dependencies, IAM permissions, data security, and monitoring.

Before implementation, I would verify the application's dependencies, runtime, database connections, and scheduled tasks. I would compare report outputs and test API performance during a two-week parallel run. The main assumption is that neither workload requires special operating-system access or persistent local storage. Any hidden dependencies or workload limits should be escalated before migration. If the report exceeds Lambda's limits, ECS on Fargate would be the fallback.

SECTION 4 | HARBORTECH OPERATIONS PLAYBOOK

Direct GitHub URL | week06-containers-serverless.md

https://github.com/Celam444453/Harbortech-playbook/blob/main/week06-containers-serverless.pdf

PLAYBOOK CONFIRMATION

Camden confirms that week06-containers-serverless.md has been published to a public GitHub repository and is accessible at the URL above.